Bachelor Thesis Summary

Simulation of LiteBIRD Instrumental Noise with the LBS Framework

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Scientific Context and Objectives

LiteBIRD (LiteB-mode polarization and Inflation from cosmic background Radiation Detection) is a space mission aimed at detecting primordial B-mode polarization in the Cosmic Microwave Background (CMB). This would prove the existence of cosmic inflation by capturing signals from primordial gravitational waves. LiteBIRD targets both the recombination peak ($\ell \simeq 80$) and the reionization peak ($\ell \lesssim 10$). Access to these large angular scales is only possible from space, but measurements are highly sensitive to low-frequency correlated noise and long-term instrumental drifts.

The primary objective of this thesis is the modeling, implementation, and comparison of instrumental noise models, in particular white noise and correlated $1/f$ noise, with the LiteBIRD Simulation (LBS) framework. To support long-duration mission simulations, I introduce a block-wise processing architecture capable of handling large amounts of data with high memory efficiency.

Noise Modeling and TOD Management

Instrumental noise is characterized by two components:

- White Noise: a stochastic process with a flat Power Spectral Density (PSD), $P(f) = \sigma^2$.
- 1/f Noise: a correlated component that exhibits low-frequency drifts, with a PSD following $P(f) \approx \sigma^2 (f_{knee}/f)^{\alpha}$.

In these models, σ^2 represents the white noise level and f_{knee} the knee frequency.

The detector output is stored as Time-Ordered Data (TOD), a 2D array ($N_{detectors} \times M_{samples}$). As M grows linearly with observation time, full-mission TODs quickly exceed the memory capacity. In this thesis I implement a block-wise approach, splitting the TOD into finite-duration segments. This ensures memory efficiency, scalability, and enables parallelization.

Implementation: LBS vs. ducc0

I developed two distinct functions to perform in-place noise addition, both based on pre-existing noise-generation functions. The first, `add_lbsNoise`, is integrated into the LBS (LiteBIRD Simulation) framework, the official Python-based library of the collaboration. The second, `add_OofaNoise`, utilizes the ducc0 (Distinctly Useful Code Collection) library, a high-performance C++ package designed for efficient astronomical data processing.

The two functions are implemented as follows:

- `add_lbsNoise`: utilizes the pre-existing `lbs.add_noise` function, which relies on FFT-based stochastic generation. While it offers high spectral flexibility (arbitrary PSDs), it requires larger memory buffers and can introduce discontinuities at block boundaries.
- `add_OofaNoise`: relies on the `ducc0.misc.OofaNoise` class, which employs recursive digital filtering. It provides intrinsic temporal continuity across blocks and a minimal, constant memory footprint, though it is constrained to a rigid $1/f^{\alpha}$ model.

Time and Frequency Domains Analysis

First, I validated my code using short-term simulations for statistical confirmation of the models. In the time domain, $1/f$ noise correctly manifests as coherent, slow-varying waves. In the frequency domain, the estimated PSDs follow the theoretical trend: the high-frequency white noise plateau and the $f^{-\alpha}$ slope below the knee frequency.

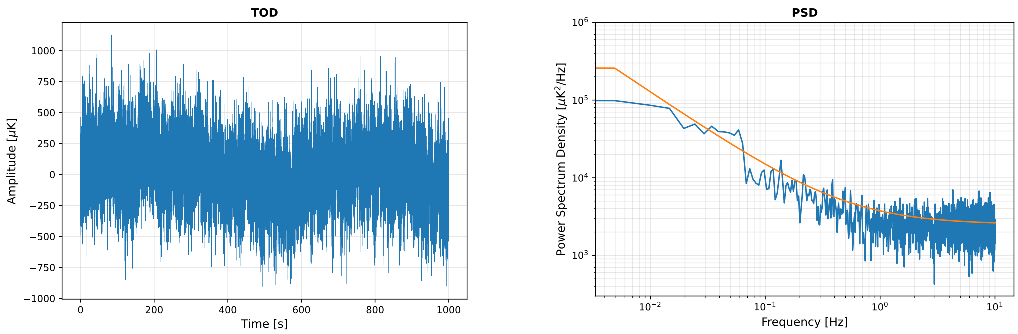

Figure 1: Statistical validation of the noise models. Left: Time-Ordered Data (TOD) showing the characteristic low-frequency drifts of $1/f$ noise over a 1000s simulation. Right: Power Spectral Density (PSD) estimate showing the transition from the $f^{-\alpha}$ correlated regime to the white-noise plateau.

Map-Making and Harmonic Analysis

To assess the spatial impact of noise, I conducted long-term simulations (up to one year) using a map-making pipeline, which uses JIT functions for optimization.

The resulting maps show typical stripping patterns. These are the result of temporal $1/f$ correlations along the telescope's scanning paths. I confirmed that the scanning strategy's redundancy helps reduce these drifts through the crossing of different scans. This effect is particularly evident in regions with high hit-counts where the noise pattern appears more attenuated.

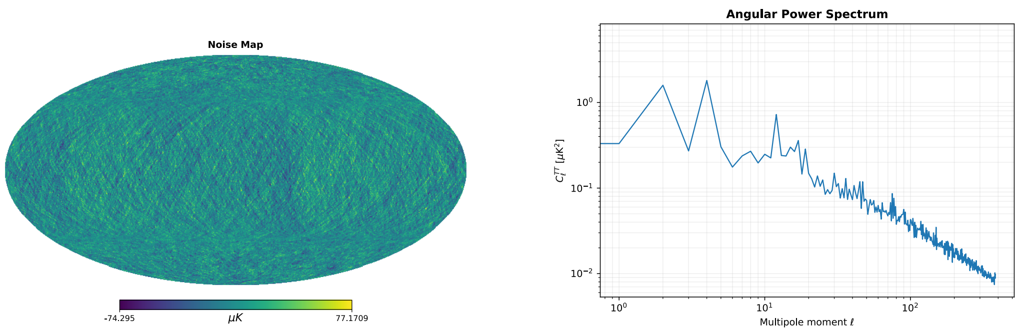

Figure 2: Spatial manifestation of instrumental noise. Left: HEALPix noise-only map generated from a one-year simulation, where the striation patterns reveal the projection of temporal correlations along the scanning strategy. Right: Angular power spectrum C_ℓ^{TT} derived from the map, assuming no correction for $1/f$ noise in the map-making. The steep rise at low multipoles highlights the dominance of correlated noise on large angular scales. This low-multipole noise is usually cleaned at the map-level in data reduction pipelines; here we preserved it to highlight its potential impact on the power spectrum if not properly treated.

Finally, I characterized the noise in the harmonic domain (C_ℓ^{TT}). Both models show a consistent descending slope at low multipoles (ℓ), representing the large-scale spatial correlations induced by $1/f$ noise. Due to the high f_{knee} value used in the tests, this correlated component dominates the spectrum, masking the white noise plateau.

Conclusions

In my thesis, I demonstrated that both implemented functions produce valid results and correctly reproduce the expected noise physics. While `add_lbsNoise` uses a standard, verified FFT-based method widely used and serves as a versatile tool for non-standard spectra, `add_OofaNoise` offers a more robust and memory-efficient solution for large-scale mission simulations. Most importantly, the implementation of the block-wise processing framework proves to be a computational necessity, enabling the simulation of realistic TOD durations.